

Exact Squarefree Orbit-Sum Reconstruction of Four- and Five-Vertex Loopless Digraphs: Minimum Separating Fingerprints, Tree Separators, and a Certified Atlas

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Abstract

Let G be S_n or A_n , acting by vertex relabelling on Boolean subsets of the $n(n-1)$ directed non-loop arcs. A squarefree orbit-sum coordinate counts, inside a directed graph, the members of one G -orbit of nonempty arc supports. For $(n, G) = (4, S_4), (4, A_4), (5, S_5), (5, A_5)$, we prove that the least degrees at which the complete cumulative coordinate dictionaries separate all graph orbits are respectively 3, 3, 4, 4. The corresponding dictionaries contain 19, 29, 83, 124 coordinates, and their exact minimum separating subfamilies have sizes 7, 11, 14, 17.

The four-vertex lower bounds decompose into one forced linear coordinate, a complete-graph vertex-cover block, and disjoint cubic constraints. The five-vertex lower bounds are certified by complete branch DAGs with 63,589 and 213,648 locally checked nodes, while explicit fingerprints attain equality. At the degree immediately below reconstruction, every residual pair for S_4 , A_4 , and S_5 has a tree-supported separator. For A_5 , 105 of 113 pairs do; the remaining eight are precisely bounded index-two split pairs. We also distinguish these global minima from the conditional problem in which all lower-degree coordinates are retained. All statements are restricted to $n = 4, 5$ and to the declared Boolean, loopless, directed, squarefree model.

Keywords: finite group actions; directed graphs; orbit sums; separating invariants; reconstruction; exact hitting sets; certified atlases.

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1 Problem, exact contribution, and boundaries

Fix $n \in \{4, 5\}$ and let G be S_n or A_n . The carrier is the set of loopless directed graphs on the labelled vertex set $[n]$, represented by Boolean arc masks. The group acts by relabelling vertices. For every G -orbit of nonempty squarefree arc supports we use the associated orbit sum as an integer-valued counting coordinate. Two finite questions are then separated:

1. what is the least support degree at which the complete cumulative dictionary distinguishes every G -orbit of graphs;
2. inside that least-degree separating dictionary, what is the least number of coordinates that still distinguishes every orbit?

The second question is global: every coordinate through the exact degree may be retained or omitted. It is not the conditional problem in which all lower-degree coordinates are fixed and only top-degree coordinates are minimized. Section 7 treats that distinction explicitly.

Write $\rho(n, G)$ for the least separating degree and $\tau(n, G)$ for the minimum cardinality inside the full dictionary through degree $\rho(n, G)$. The central finite statement is the following.

Theorem 1.1 (Unified four- and five-vertex reconstruction theorem). *For*

$$(n, G) = (4, S_4), (4, A_4), (5, S_5), (5, A_5),$$

the exact least separating degrees are respectively

$$\rho(n, G) = 3, 3, 4, 4.$$

The full dictionaries at those degrees contain respectively 19, 29, 83, 124 coordinates, and their minimum separating subfamilies have cardinalities

$$\tau(n, G) = 7, 11, 14, 17.$$

Proof. The four-vertex degree and minimum assertions are proved in Section 4: explicit degree-two collisions give necessity, complete degree-three signature tables give sufficiency, certified supports attain 7 and 11, and disjoint structural constraints give matching lower bounds. The five-vertex assertions are proved in Section 5: the residual degree-three pairs give necessity, the full quartic signatures give sufficiency, witnesses of sizes 14 and 17 give upper bounds, and complete checked branch DAGs exclude budgets 13 and 16. The four cases use the same definitions in Sections 2 and 3. $\square$

Table 1: The exact finite panel. Coordinate profiles give the number of support orbits in each exact degree, starting at degree one.

Action	pattern orbits	coordinate profile	dictionary	degree	minimum
S_4	218	1/5/13	19	3	7
A_4	368	1/7/21	29	3	11
S_5	9,608	1/5/16/61	83	4	14
A_5	17,824	1/6/21/96	124	4	17

The theorem is accompanied by a support-shape statement. One degree below exact reconstruction, every residual pair for S_4 , A_4 , and S_5 has an individual tree-supported separator. For A_5 , exactly eight residual pairs are tree-blind, and each is the pair of A_5 -orbits inside one S_5 -orbit. Section 6 records the exact finite meaning of that index-two contrast.

Scope boundary. The objects are binary, directed, and loopless. Coordinates are squarefree orbit sums of nonempty arc supports. Equivalence is the stated vertex action, not complementation, transposition, or an external quotient. The only vertex counts are four and five. The paper does not infer a formula for other vertex counts, does not replace the squarefree dictionary by a full polynomial invariant space, and gives the index-two terminology no physical interpretation.

The certified atlas $\mathcal{A}$ is part of the evidentiary package. It contains canonical registries, collision masks, upper witnesses, lower certificates, and independent reconstructions. The paper and atlas have different jobs: the paper states and explains the mathematics, while the atlas makes each bounded enumeration and optimization replayable.

2 Loopless directed carrier, actions, and squarefree dictionaries

Let $V = [n] = \{0, \dots, n-1\}$ and

$$\Omega_n = \{(i, j) \in V^2 : i \neq j\}.$$

A labelled loopless digraph is a subset $P \subseteq \Omega_n$, or equivalently its indicator $x^P \in \{0, 1\}^{\Omega_n}$. The selected group $G \leq S_n$ acts by

$$g \cdot (i, j) = (g(i), g(j)), \quad g \cdot P = \{g \cdot a : a \in P\}.$$

The objects to be reconstructed are the orbits $G \backslash \{0, 1\}^{\Omega_n}$. There are 218, 368, 9608, 17824 of them for the four actions in Table 1.

For a nonempty support $T \subseteq \Omega_n$, set

$$m_T(x) = \prod_{a \in T} x_a, \quad q_{[T]}^G(x) = \sum_{U \in G \cdot T} m_U(x),$$

where the sum is over the distinct supports in the orbit. On a Boolean pattern P this specializes to

$$q_{[T]}^G(P) = \#\{U \in G \cdot T : U \subseteq P\}. \quad (1)$$

Thus every coordinate is an integer count. It depends only on the orbit of P and on the support orbit of T .

Definition 2.1 (Cumulative squarefree dictionary). For $d \geq 1$, let

$$\mathcal{D}_{n, \leq d}^G = \{q_{[T]}^G : \emptyset \neq T \subseteq \Omega_n, |T| \leq d\},$$

with one coordinate for each G -orbit of supports. Its signature map is

$$\Phi_{n, \leq d}^G([P]_G) = \left(q(P) \right)_{q \in \mathcal{D}_{n, \leq d}^G}.$$

The reconstruction degree $\rho(n, G)$ is the least d for which this map is injective.

The exact-degree coordinate profiles are

G	$(\mathcal{D}_{n,1}^G , \mathcal{D}_{n,2}^G , \dots)$
S_4	(1, 5, 13)
A_4	(1, 7, 21)
S_5	(1, 5, 16, 61)
A_5	(1, 6, 21, 96).

The sums of these rows give the dictionary sizes in the central theorem. In particular, a profile entry is an exact-degree count, not a cumulative count.

2.1 Canonical finite encoding

The atlas orders arcs lexicographically by (i, j) with $i \neq j$. A pattern or support is serialized by the corresponding unsigned bit mask. Its canonical representative is the minimum unsigned mask in its G -orbit. Coordinates are ordered by degree and subsequently by canonical support mask. The identifier therefore contains the action, vertex count, degree, and canonical mask. For four vertices, compact table identifiers are resolved through the coordinate registry; the five-vertex registry uses the mask directly. Both forms resolve to a unique canonical support.

This encoding is convenient but not mathematical content: replacing it by any bijective serialization leaves all orbit sums, signatures, and separator masks unchanged. Canonicalization merely permits byte-level comparison of two independent reconstructions.

2.2 Orbit incidence and the normalization lock

For a pattern orbit R and support orbit C , let

$$A_{R,C} = \#\{T \in C : T \subseteq P\} \quad (P \in R)$$

and

$$M_{R,C} = \#\{P \in R : T \subseteq P\} \quad (T \in C).$$

Both quantities are well defined by transitivity. Double-counting incident pairs $(P, T) \in R \times C$ gives

$$|R|A_{R,C} = |C|M_{R,C}. \tag{2}$$

The signature entries in this paper are the $A_{R,C}$ values of Equation (1). Equation (2) is essential: the two incidence normalizations are not identified unless their orbit-size factors are carried explicitly. The matrix vocabulary is related to classical orbit-incidence constructions such as the Kramer–Mesner matrix [KM76], but none of the finite values below is imported from that literature.

Orbit enumeration and orbit sums sit in a broad classical framework [Pól37, Mnu92, RS06]. Here the carrier, group, squarefree support restriction, and integer normalization are fixed before any separation or optimization is performed.

3 Exact separation as a hitting-set problem

Fix one of the four actions and a finite coordinate dictionary $\mathcal{D}$. For distinct pattern orbits R, R' , define the separator mask

$$\text{Sep}_{\mathcal{D}}(R, R') = \{q \in \mathcal{D} : q(R) \neq q(R')\}.$$

The full dictionary separates if and only if every such mask is nonempty. A subfamily $F \subseteq \mathcal{D}$ separates if and only if

$$F \cap \text{Sep}_{\mathcal{D}}(R, R') \neq \emptyset \quad \text{for every unordered pair } \{R, R'\}. \quad (3)$$

Consequently the minimum fingerprint problem is the transversal-number problem for the hypergraph of nonempty pair-separator masks.

Each coordinate can take several integer values. For this reason our primary language is “hitting set on unordered orbit-pair separation masks.” The classical Test Cover problem provides useful optimization context [BHH⁺03], but applying that name here requires the explicit multi-outcome generalization encoded by Equation (3).

The reported minima count selected coordinates from the declared dictionary. They are not minima among arbitrary integer-valued combinations of those coordinates. Indeed, on a finite pattern set, a sufficiently large mixed-radix linear combination can encode the entire separating signature in one integer whenever the full dictionary separates. The nontrivial optimization problem here is therefore the sparsest separating *subfamily* of the fixed coordinate dictionary.

3.1 Lossless reductions

Two elementary reductions shrink the constraint family without changing its transversals. Duplicate separator masks may initially be represented once. Second, if $S \subsetneq T$ are separator masks, the constraint “hit T ” is implied by “hit S ” and T may be deleted. The resulting inclusion-minimal antichain is the finite hitting-set instance used for the four-vertex readable proof. The five-vertex lower certificate instead uses a selected explicit pair core; infeasibility of that core at a given budget implies infeasibility of the complete pair system at the same budget.

3.2 Upper and lower certificates

An upper certificate is an explicit list of coordinates. Its checker recomputes every value in Equation (1) and verifies that all pattern-orbit signature rows are distinct. Cardinality then gives the upper bound.

A lower certificate fixes a budget b and proves that no subfamily of size at most b hits all required masks. Three locally checkable rules suffice:

1. a singleton separator mask forces its sole coordinate;
2. k pairwise disjoint remaining masks require at least k coordinates;
3. if every coordinate hits at most M among C remaining masks, at least $\lceil C/M \rceil$ coordinates are required.

If none of these closes a state, branch on one uncovered mask. Every hitting set must select at least one of its coordinates, so the children are jointly exhaustive. Memoizing identical reduced states turns the tree into a DAG. Appendix C gives the precise checker contract.

The proof semantics therefore do not depend on an optimizer’s status code. Optimization software was used to discover candidate supports and hard pair cores for the five-vertex instances. The accepted lower bounds are the complete DAGs checked from explicit pattern pairs and recomputed separator masks.

3.3 Global versus residual pair systems

Two pair systems occur in later sections. The *global* minimum uses all unordered pairs and every coordinate through the exact degree. The *top-degree extension* problem retains every lower-degree coordinate; only pairs still equal on those coordinates need to be hit by top-degree coordinates. Confusing these universes changes the objective; the exact conditional values are stated separately in Section 7.

4 The four-vertex exact degree and minimum fingerprints

There are twelve loopless directed arcs on four labelled vertices, hence $2^{12} = 4096$ labelled patterns. Canonicalization gives 218 orbits under S_4 and 368 under A_4 . The symmetric count also agrees with the independent unlabelled-digraph census [McK26]; every load-bearing signature and minimum assertion is checked by the distributed certificate verifier.

Proposition 4.1 (Four-vertex collision boundary). *For S_4 , the signature through degree two has 34 non-singleton fibres covering 70 pattern orbits, with maximum fibre size four and 39 unordered residual pairs. For A_4 , it has 51 non-singleton fibres covering 102 pattern orbits, all of size two and hence 51 residual pairs. Adding all degree-three coordinates separates all 218 and 368 pattern orbits.*

Proof. Enumerate all labelled masks, replace each by the minimum mask in its group orbit, and evaluate Equation (1) on the canonical representatives. Grouping equal rows through degree two gives the stated fibre census. The non-singleton fibres explicitly witness failure at degree two. Appending all cubic columns yields respectively 218 and 368 distinct rows, equal to the orbit counts, so the cumulative degree-three signature is injective. Two separate implementations agree on the canonical pattern registry, support registry, integer signature matrix, fibres, and cubic separator masks; an independent verifier recomputes injectivity rather than trusting the recorded census. $\square$

It follows immediately that $\rho(4, S_4) = \rho(4, A_4) = 3$. The full dictionaries contain $1 + 5 + 13 = 19$ and $1 + 7 + 21 = 29$ coordinates.

Theorem 4.2 (Four-vertex global minimum fingerprints). *Within the complete degree-at-most-three dictionaries, the exact minimum separating sizes are 7 of 19 for S_4 and 11 of 29 for A_4 . The lower bounds are the sums of a forced degree-one coordinate, a complete-graph vertex-cover block on the quadratic coordinates, and respectively two or four coordinate-disjoint cubic constraints.*

Proof. For the upper bounds, direct evaluation verifies the canonical supports

$$F_{S_4} = \{S4-D1-C000, S4-D2-C000, S4-D2-C001, S4-D2-C002, S4-D2-C003, \\ S4-D3-C002, S4-D3-C011\},$$

$$F_{A_4} = \{A4-D1-C000, A4-D2-C000, A4-D2-C001, A4-D2-C002, A4-D2-C003, \\ A4-D2-C004, A4-D2-C005, A4-D3-C000, A4-D3-C010, \\ A4-D3-C011, A4-D3-C014\}.$$

They separate all $23,653 = \binom{218}{2}$ and $67,528 = \binom{368}{2}$ unordered pairs.

For the lower bounds, compare the empty pattern with a one-arc pattern. Their separator mask is the unique linear coordinate, so that coordinate is forced. Next let T_i, T_j represent two distinct quadratic support orbits. The patterns consisting exactly of T_i and T_j have equal linear count, every cubic coordinate vanishes on both, and their quadratic separator mask is exactly $\{q_i, q_j\}$. Thus every pair of quadratic coordinates is an edge constraint. A separating family contains a vertex cover of K_5 for S_4 or K_7 for A_4 , requiring at least four or six quadratic coordinates.

Finally, the certified inclusion-minimal mask antichains contain two pairwise coordinate-disjoint pure-cubic constraints for S_4 and four for A_4 . They require at least two or four cubics. The three contributions lie in disjoint degree layers, giving

$$1 + 4 + 2 = 7, \quad 1 + 6 + 4 = 11.$$

The displayed families attain both bounds. Appendix B records the finite cubic blocks and the uniform branch-DAG cross-check. $\square$

Table 2: Four-vertex global hitting-set instances and readable lower-bound decomposition. The three components occupy disjoint degree layers.

Action	full pairs	minimal masks	forced	cover	cubic	total
S_4	23,653	26	1	$\tau(K_5) = 4$	2	7
A_4	67,528	58	1	$\tau(K_7) = 6$	4	11

For auditability, the complete pair system reduces initially to 6998 and 40,786 distinct nonempty separator masks, then to inclusion-minimal antichains of sizes 26 and 58. A second lower proof, independent of the readable decomposition, is a complete branch DAG with 11 nodes for S_4 and 22 for A_4 , excluding budgets six and ten. These small DAGs and the structural proof arrive at the same lower bounds by different presentations.

5 The five-vertex exact degree and minimum fingerprints

On five labelled vertices there are twenty non-loop arcs and 2^{20} labelled patterns. Canonicalization gives 9608 S_5 -orbits and 17,824 A_5 -orbits. The symmetric count agrees with McKay's independent unlabelled-digraph census [McK26]; all load-bearing signatures

and minima are checked by the distributed certificate verifier. The exact-degree coordinate profiles are (1, 5, 16, 61) and (1, 6, 21, 96), so the cumulative quartic dictionaries have sizes 83 and 124.

Theorem 5.1 (Five-vertex degree and global minimum fingerprints). *The signatures through degree three leave 190 unordered residual pairs for S_5 and 113 for A_5 . The signatures through degree four separate all 9608 and 17,824 pattern orbits, so the least separating degree is four for both actions. Within the full degree-at-most-four dictionaries, the exact minimum separating sizes are 14 of 83 and 17 of 124. Explicit fingerprints attain those sizes, while complete checked branch DAGs exclude budgets 13 and 16.*

Proof. The nonempty residual-pair lists certify noninjectivity through degree three. For degree four, independently reconstructed signature matrices have one row for every canonical pattern orbit and no repeated row. This proves injectivity and hence exact degree four.

For the global upper bounds, the checker evaluates all coordinates on all canonical pattern representatives and verifies injectivity for the following families. The canonical coordinate IDs are grouped by degree for readability:

S_5 , cardinality 14.

S5:5:1:00001;
S5:5:2:00003, S5:5:2:00011, S5:5:2:00012, S5:5:2:00022;
S5:5:3:00016, S5:5:3:00025, S5:5:3:00148, S5:5:3:00448;
S5:5:4:0006a, S5:5:4:00125, S5:5:4:00145, S5:5:4:00158, S5:5:4:00164.

A_5 , cardinality 17.

A5:5:1:00001;
A5:5:2:00011, A5:5:2:00012, A5:5:2:00022,
A5:5:2:00024, A5:5:2:00028;
A5:5:3:00016, A5:5:3:0002a, A5:5:3:0002c,
A5:5:3:00124, A5:5:3:00144, A5:5:3:00184,
A5:5:3:00188;
A5:5:4:00116, A5:5:4:00198, A5:5:4:001c4, A5:5:4:01182.

For the lower bounds, use explicit pair cores of 249 and 285 separator masks. Each row stores both canonical pattern IDs, not merely a precomputed mask. The checker reconstructs the group, coordinate orbits, pattern representatives, integer evaluations, and every core separator mask. It then checks a complete branch DAG rooted at budget 13 or 16. At a branch node, every possible choice from a named uncovered separator mask appears as a child after forced singleton propagation. Each leaf is closed either by a packing of more pairwise disjoint masks than the remaining budget or by the uniform maximum-coverage bound. All children and all leaves are therefore locally exhaustive. The S_5 DAG has 63,589 nodes and depth 11; the A_5 DAG has 213,648 nodes and depth 14. Induction from the leaves to the roots proves infeasibility at the two budgets. Together with the upper families, this gives minima 14 and 17. $\square$

Table 3: Five-vertex complete lower-bound DAGs. Each root budget is one below the displayed certified minimum.

Action	coordinates	core pairs	budget	nodes	depth	minimum
S_5	83	249	13	63,589	11	14
A_5	124	285	16	213,648	14	17

The accepted proof objects distinguish discovery from certification. Mixed-integer optimization was useful for locating the witnesses and the pair cores. Its infeasibility messages are retained only as diagnostics. The lower-bound authority is the standard-library checker that reconstructs the mathematics and validates every DAG transition. A normal run and a run with Python assertions disabled produce the same scientific verdict.

The two DAGs contain 277,237 nodes in total. Their canonical uncompressed JSON representations are about 40 MB and 146 MB; deterministic gzip transport reduces them to about 5 MB and 19 MB. Compression and hashes make transport manageable, while the checker supplies the semantic proof.

6 Tree separators and the bounded index-two contrast

A top-degree support has size $n - 1$: cubic for $n = 4$ and quartic for $n = 5$. Call such a directed support a *tree support* if the simple undirected graph obtained by forgetting arc orientations is a tree on all n vertices. This is a property of the support orbit and hence defines a subset of the top-degree coordinate dictionary.

For $A_n \triangleleft S_n$ of index two, an S_n -orbit is either one A_n -orbit or the disjoint union of two A_n -orbits. In the second case we call the resulting pair an *index-two split pair*. This is purely a finite group-action definition.

Proposition 6.1 (Tree and index-two contrast in the finite panel). *Every one of the 39 S_4 and 51 A_4 residual pairs at degree two, and every one of the 190 S_5 residual pairs at degree three, has an individual tree-supported separator at the exact top degree. For A_5 , 105 of the 113 degree-three residual pairs have such a separator. The remaining eight pairs are tree-blind, and each of those eight is an index-two split pair of A_5 -orbits contained in one S_5 -orbit.*

Proof. For every residual pair, compute its separator mask inside the exact top-degree dictionary and intersect that mask with the coordinates whose canonical supports satisfy the tree test. The intersections have the stated nonempty/empty census. This uses all 8, 12, 27, 41 tree-supported top-degree coordinates for S_4 , A_4 , S_5 , A_5 , respectively.

Independently, construct the map from each A_n pattern orbit to the containing S_n orbit. A fibre of size two is an index-two split. There are nine split residual pairs for A_4 and fifty-four for A_5 . Every A_4 split pair is tree-separated. In the A_5 residual list, all eight empty tree intersections occur among the fifty-four split pairs; exact pattern IDs and separator masks are stored in the atlas. Rebuilding the full payload by a second implementation gives byte-identical canonical data, including this crosswalk. $\square$

Table 4: Residual unordered pairs one degree below exact reconstruction and their top-degree tree-separator census. “Split pairs” are pairs of A_n -orbits contained in one S_n -orbit.

Action	residual degree	residual pairs	tree-separable	split pairs
S_4	2	39	39	0
A_4	2	51	51	9
S_5	3	190	190	0
A_5	3	113	105	54

The converse implication is false in the finite data: forty-six of the fifty-four A_5 split residual pairs do have a tree separator. Thus index-two splitting is necessary for the observed tree blindness but does not by itself force it. This distinction prevents the eight exceptions from being inflated into a general statement about alternating actions.

6.1 A common tree cover for the symmetric five-vertex action

The proposition concerns one separator per residual pair. A stronger finite optimization asks for a single family of tree coordinates that hits all 190 S_5 residual pairs. Its exact minimum is five, and there are 88 minimum tree-only families in the canonical coordinate order. For A_5 no tree-only family can hit every residual pair because the eight tree-blind pairs have empty tree separator masks.

When the tree restriction is removed, each of the two minimum conditional S_5 quartic extensions has four coordinates: exactly three have tree supports and the fourth has an underlying support that is unicyclic on its touched vertices. This shape statement belongs to the bounded atlas and is not used in the proof of the global 14-coordinate minimum.

Scope boundary. The words “split pair” and “tree-blind” refer only to orbit fibres and directed support shapes in the declared finite model. They carry no geometric, stereochemical, or physical assertion.

7 Conditional top-degree extensions versus global minima

Let $d = \rho(n, G)$ and retain the entire lower dictionary $\mathcal{D}_{n, \leq d-1}^G$. A *conditional top-degree extension* is a set $H \subseteq \mathcal{D}_{n, d}^G$ for which

$$\mathcal{D}_{n, \leq d-1}^G \cup H$$

is separating. Equivalently, H hits only the residual-pair separator masks left by the complete lower dictionary. This differs from the global problem, where lower-degree coordinates may also be omitted.

Proposition 7.1 (Exact conditional top-degree extensions). *When every lower-degree coordinate is retained, the minimum number of added top-degree coordinates and the number of optimal subsets are*

$$(2, 2), (4, 128), (4, 2), (4, 200)$$

for S_4, A_4, S_5, A_5 , respectively. For the five-vertex actions, the two and two hundred optimal subsets form respectively one and fifty orbits under the recorded external diagnostic transformations. These quotient counts are not part of the scientific equivalence relation. The four conditional minima are distinct from the global minima 7, 11, 14, 17.

Proof. For each action, restrict every residual-pair separator mask to the exact top-degree coordinates. An exhaustive minimum hitting-set computation gives the stated minima and complete optimal-subset census. Independent implementations reconstruct the residual pairs and separation masks and agree on every optimal subset. Direct signature evaluation checks each recorded upper family; exhaustive exclusion below the minimum supplies the lower side.

For S_5 and A_5 , apply the separately recorded transformations only after the optimal subsets have been determined. Their orbits have sizes compatible with the totals 2 and 200, yielding diagnostic counts 1 and 50. Because these transformations are not added to S_5 or A_5 , quotienting does not change the graph-orbit equivalence or the optimization result. $\square$

Table 5: Conditional top-degree extensions. All lower-degree coordinates are retained. The diagnostic quotient counts are not part of the scientific equivalence relation.

Action	lower dictionary	added minimum	total	optima	quotient orbits
S_4	6	2	8	2	—
A_4	8	4	12	128	—
S_5	22	4	26	2	1
A_5	28	4	32	200	50

The resulting total fingerprint sizes are 8, 12, 26, 32 because the retained lower dictionaries contain 6, 8, 22, 28 coordinates. These totals can exceed the global minima substantially: in the global problem the optimizer may discard lower-degree coordinates whose separating work is replicated by the chosen top-degree family. In particular, the former greedy bounds 8 and 12 for the four-vertex actions become exact only for this conditional problem, not for the global 19- and 29-coordinate hitting sets.

8 Certificates and independent replay

The certified atlas $\mathcal{A}$ is a collection of finite mathematical objects with an explicit dependency graph. It is not a substitute for the statements and proof explanations in Sections 4–7. Its role is to expose every enumerated carrier, coordinate, collision, witness, and lower-bound transition to a checker.

8.1 Traceability from statement to certificate

The six main statements have the following evidence chain:

Each principal surface is addressed by repository-relative path and SHA-256 in Appendix D. The separate exact crosswalk also records statement text, support files, verifier, and transfer limit. This prevents a typeset number from becoming detached from the finite object that establishes it.

Table 6: Division of labour between mathematical statements and finite certificate surfaces.

Surface	Mathematical job	Principal finite object
Carrier and dictionaries	Fix the objects, action, support orbits, and integer evaluation	canonical pattern and coordinate registries
Degree exactness	Exhibit a collision below the bound and injectivity at the bound	collision fibres and full signature matrices
Upper minimum bound	Give a separating coordinate family of the stated size	explicit coordinate IDs plus direct injectivity check
Lower minimum bound	Exclude every smaller family	structural blocks for $n = 4$; complete branch DAGs for $n = 5$
Tree/index-two contrast	Classify top-degree support shape and split pairs	separator masks, tree flags, and S_n/A_n crosswalk
Transport and replay	Bind bytes without treating hashes as semantic truth	manifest, canonical encodings, and independent checkers

Statement	finite content	principal certificate surface
Unified reconstruction theorem	degrees 3, 3, 4, 4 and minima 7, 11, 14, 17	union of the four exact-degree and global-minimum packages
Four-vertex collision proposition	39/51 residual pairs and cubic sufficiency	collision atlas and degree-three signature tables
Four-vertex minimum theorem	global minima 7/19 and 11/29	explicit supports, structural blocks, and 11/22-node lower DAGs
Five-vertex degree and minimum theorem	exact degree four and minima 14/83, 17/124	complete signatures, upper witnesses, and 63,589/213,648-node lower DAGs
Tree/index-two proposition	39, 51, 190, 105 tree-separable residual pairs	tree flags, separator masks, and S_n/A_n orbit crosswalk
Conditional-extension proposition	added-coordinate data (2, 2), (4, 128), (4, 2), (4, 200)	residual-pair payloads and complete optimal-subset census

Table 7: Statement-to-certificate traceability.

8.2 Independent implementation paths

The five-vertex computation uses separate producer and checker code trees. The producer generated groups by direct permutation enumeration and evaluated support orbits by its own incidence path. The checker generated groups by closure from generators, reconstructed support orbits independently, and evaluated coordinates by contained-support accumulation. Neither imported the other’s module or trusted a producer-generated separator mask. The two complete canonical payloads are byte-identical.

For the lower proofs, the checker reconstructs all 534 core pattern pairs, all coordinate IDs, and every separator mask before reading DAG transitions. It then checks reachability, acyclicity, every branch child, and every packing or coverage leaf. Thus neither a stored separator mask nor an optimizer status is accepted as proof authority.

The assurance label is deliberately limited: the implementations are independent in implementation and data path. This is not a claim of blind human replication or independent model-family authorship.

8.3 Hashes and semantic checks

A hash authenticates that the checked byte string is the same byte string later inspected. It does not show that the string encodes the intended group, coordinate evaluation, or proof rule. Those assertions are verified by reconstruction. The atlas therefore stores transport evidence (the manifest and hashes) together with semantic evidence (explicit patterns, supports, integer values, and local branch rules). No hash-only oracle is accepted for a mathematical result.

9 Prior-art subtraction, terminology, and limitations

The source map assigns each reference a typed role and a transfer limit. The external literature supplies general frameworks, vocabulary, and adjacent problems; the exact finite values in this paper are supported by the distributed certificates. The distinction is summarized below.

Role and sources	What transfers	What does not transfer
Orbit enumeration [P6l37]	finite group-action enumeration framework	injectivity and minimum separator cardinality
Separating invariants [Kem09, DJ15, KLR22]	general separation language and constraints	minima in the fixed integer squarefree dictionaries
Reconstruction and decks [Mnu92, Sim18, DJK ⁺ 26]	orbit-sum and reconstruction context	equality with the directed Boolean carrier or its signatures
Graph-counting coordinates [Thi00, Mik08, CDM17, AFKV20, BHR ⁺ 26]	adjacent graph-invariant and motif-transform context	the exact directed fibres, residual pairs, or minima

Role and sources	What transfers	What does not transfer
Orbit-incidence matrices [KM76]	matrix vocabulary	an unnormalized identification of the two incidence conventions
Test Cover [BHH ⁺ 03]	classical optimization context	a binary-test formulation without the multi-outcome caveat
Representation and graph isomorphism [GU26]	adjacent support-degree context	any theorem used to derive the four finite numerical rows

The graph-reconstruction work of Dufresne, Jeronimo, Kenkel, Lindo, and Villamizar [DJK⁺26] concerns an undirected vertex-deletion setting and supplies boundary context, not an identification of carriers. Similarly, general reconstruction under group actions [RS06] does not determine the selected squarefree coordinates or their finite hitting-set minima.

The source audit is bounded and typed. Failure to identify an exact match in that corpus is not used as a mathematical premise. All theorems in this paper remain true or false solely by their stated finite definitions and certificates; bibliographic scope affects contextual positioning, not the validity of the enumerations.

9.1 Model limitations

The conclusions depend on every item in the following list:

- the vertex set has size four or five;
- arcs are directed, loops are excluded, and multiplicities are Boolean;
- the acting group is exactly S_n or A_n by vertex relabelling;
- coordinates are orbit sums of nonempty squarefree arc supports;
- the dictionaries are cumulative through the declared support degree;
- minimum cardinality is taken among coordinate subfamilies, not among arbitrary formulas generated from them;
- transposition, complementation, and odd external relabellings are controls or diagnostic transformations, not extra equivalences.

No computation in the governed experiment uses a vertex count above five. The data therefore do not support a recurrence or asymptotic law in n . Likewise, the eight A_5 tree-blind pairs establish a bounded contrast, not a general obstruction theorem for alternating groups. The source-shape census does not change the global minimum proof, and the conditional extension counts do not replace the global minima.

10 Conclusions within the four- and five-vertex boundary

The same finite reconstruction problem yields a coherent four-row theorem. For S_4, A_4, S_5, A_5 , the exact squarefree orbit-sum degrees are 3, 3, 4, 4; the least-degree separating dictionaries have sizes 19, 29, 83, 124; and their global minimum fingerprints have sizes 7, 11, 14, 17. The four-vertex lower bounds have short structural decompositions. The five-vertex lower bounds have complete portable branch DAGs and explicit equality witnesses.

Tree-supported top-degree coordinates then expose a bounded structural contrast. They individually separate every residual pair except eight in the A_5 case; each exception lies across an index-two split, while many other split pairs remain tree-separated. This sharpens the finite description without extending its domain.

The paper and atlas form one evidentiary object. Definitions and readable proof semantics are in the paper; complete registries, separator masks, witnesses, DAGs, and independent reconstructions are in the atlas. The distributed verifier reconstructs the finite semantics instead of treating stored hashes or optimizer status as proof. The resulting claims remain confined to the four declared actions and coordinate dictionaries.

A Conventions and standard finite-action lemmas

This appendix isolates the elementary facts used to convert the finite group action into signature and hitting-set statements. They are included so that the computational objects can be read against explicit mathematical semantics.

Lemma A.1 (Orbit-sum invariance and Boolean evaluation). *For $T \subseteq \Omega_n$ and $g \in G$,*

$$q_{[T]}^G(g \cdot P) = q_{[T]}^G(P).$$

On Boolean patterns, this common value is the number of supports in $G \cdot T$ contained in P .

Proof. Relabelling permutes the support orbit:

$$\begin{aligned} q_{[T]}^G(g \cdot x) &= \sum_{U \in G \cdot T} m_U(g \cdot x) = \sum_{U \in G \cdot T} m_{g^{-1}U}(x) \\ &= \sum_{W \in G \cdot T} m_W(x). \end{aligned}$$

For a Boolean pattern, $m_U(P)$ is one exactly when $U \subseteq P$ and is zero otherwise. □

Lemma A.2 (Orbit-incidence normalization). *With $A_{R,C}$ and $M_{R,C}$ as in Section 2,*

$$|R|A_{R,C} = |C|M_{R,C}.$$

Proof. Count the set

$$I_{R,C} = \{(P, T) \in R \times C : T \subseteq P\}$$

by P and then by T . Transitivity makes the two fibre sizes constant, giving $|I_{R,C}| = |R|A_{R,C} = |C|M_{R,C}$. □

Lemma A.3 (Pair-separation equivalence). *For a finite dictionary $\mathcal{D}$ and subfamily $F \subseteq \mathcal{D}$, the restricted signature Φ_F is injective on pattern orbits if and only if F intersects every nonempty mask $\text{Sep}_{\mathcal{D}}(R, R')$.*

Proof. The signatures of R and R' agree on F precisely when every coordinate in F has equal values on the two rows, which is equivalent to $F \cap \text{Sep}_{\mathcal{D}}(R, R') = \emptyset$. $\square$

Lemma A.4 (Duplicate and antichain reductions). *Let $\mathcal{S}$ be a family of nonempty subsets of a finite coordinate set. Deleting a duplicate set, or deleting $T \in \mathcal{S}$ when some $S \in \mathcal{S}$ satisfies $S \subsetneq T$, does not change the family of hitting sets.*

Proof. A duplicate imposes the same condition twice. If a family hits S , then it also hits every superset T ; hence the latter condition is redundant. $\square$

Lemma A.5 (Canonical representatives preserve the finite model). *Choosing the minimum unsigned bit mask in every pattern and support orbit induces bijections between abstract orbits and registry rows. Evaluation on one canonical pattern representative therefore determines the coordinate on the entire pattern orbit.*

Proof. Every finite orbit has a unique minimum mask. Distinct orbits are disjoint and hence have distinct minima. Orbit-sum invariance then gives the evaluation claim. $\square$

The last lemma explains why comparing canonical registries and signature matrices is stronger than comparing only aggregate counts: it aligns every abstract orbit and every coordinate before row-level comparison.

B Readable four-vertex lower-bound certificates

This appendix expands the lower-bound part of Theorem 4.2. It deliberately stops at the ingredients needed for the minima 7 and 11; it does not classify all equality families.

B.1 The forced linear layer

There is one degree-one support orbit: a single directed arc. Let P_0 be the empty pattern and P_1 a one-arc pattern. Every coordinate of degree at least two vanishes on both, while the linear orbit sum takes values zero and one. The separator mask is therefore a singleton, forcing the linear coordinate in every separating family.

B.2 The complete quadratic constraint graph

Let $T_1, \dots, T_r$ be canonical representatives of the quadratic support orbits, where $r = 5$ for S_4 and $r = 7$ for A_4 . For $i \neq j$, compare the two patterns $P_i = T_i$ and $P_j = T_j$. Both have two arcs, so their unique linear coordinate values agree. Every cubic coordinate vanishes. Among the quadratic coordinates, $q_{[T_i]}$ and $q_{[T_j]}$ are the only coordinates whose values distinguish the two support-orbit representatives. Hence

$$\text{Sep}(P_i, P_j) = \{q_{[T_i]}, q_{[T_j]}\}.$$

All $\binom{r}{2}$ pairs occur, producing the edge set of K_r . A hitting set must contain a vertex cover of that complete graph. Since $\tau(K_r) = r - 1$, at least four of the five S_4 quadratics and six of the seven A_4 quadratics are necessary.

B.3 Coordinate-disjoint cubic constraints

After the duplicate and superset reductions, the following explicit pattern pairs have pure-cubic separator masks. Within each action their coordinate sets are pairwise disjoint.

Action	pattern pair	cubic separator coordinates
S_4	P111 / P155	C001, C002, C006, C007
S_4	P052 / P066	C004, C005, C009, C011
A_4	P049 / P060	C002, C005, C011, C016
A_4	P112 / P115	C006, C009, C014, C018
A_4	P093 / P096	C001, C008, C010, C019
A_4	P106 / P153	C000, C003, C004, C007, C012, C013, C017, C020

Here C_j abbreviates the coordinate ID S_4 -D3- C_j or A_4 -D3- C_j in the corresponding row, with three decimal digits. The atlas stores the full pattern masks, values, pair IDs, and separator-mask integers. Pairwise disjointness makes the lower contribution two for S_4 and four for A_4 .

Because the linear, quadratic, and cubic coordinate sets are disjoint, their lower bounds add. This proves $1 + 4 + 2 = 7$ and $1 + 6 + 4 = 11$ without assuming that all lower-degree coordinates are retained.

B.4 Uniform branch-DAG cross-check

The complete reduced constraint antichains have 26 and 58 rows. Starting with budgets six and ten, a generic branch checker repeatedly applies forced singleton and disjoint-packing rules and branches on an uncovered constraint. The resulting proof DAGs have 11 and 22 nodes. At every branch the listed children cover every coordinate choice available in the selected constraint; every leaf displays a packing larger than its remaining budget. Independent verification of these DAGs supplies a method-uniform check of the more readable degree-layer argument above.

C Five-vertex branch-DAG encoding and checker contract

The five-vertex lower certificates prove infeasibility of a hitting-set budget on an explicit core of pair-separator masks. This appendix specifies why a locally accepted DAG certifies the global lower bound.

C.1 Root data

For each action the proof object contains the ordered coordinate IDs, the budget, and a list of explicit unordered pattern pairs. The checker rebuilds each canonical pattern from its ID and recomputes the integer coordinate rows before deriving the separator mask. The root statistics are:

G	budget	core rows	root-state SHA-256 prefix
S_5	13	249	80a7d31c66c96555
A_5	16	285	0fb0e501622ad7bc.

The pair core is a subset of the complete separator system. Therefore, if no budget- b family hits the core, no budget- b family separates all pattern orbits.

C.2 Reduced states and local rules

A state consists of a nonnegative remaining budget and a set of uncovered separator masks. Selecting coordinate i removes every mask containing i and decreases the budget by one. Singleton masks are propagated until none remains. The state is rejected if the budget is negative or a remaining mask is empty.

A *branch node* names one uncovered mask S and contains one child for every $i \in S$, after selecting i and applying deterministic singleton propagation. Since a hitting set must meet S , these children exhaust all solutions below that node.

A *packing leaf* supplies k pairwise disjoint uncovered masks when k exceeds the remaining budget. Each selected coordinate hits at most one packing member, so closure is valid. A *coverage leaf* supplies the number C of uncovered masks and an exact maximum M of masks hit by any remaining coordinate. If $\lceil C/M \rceil$ exceeds the remaining budget, closure is again valid.

Every state stores a canonical state hash, but the checker recomputes the reduced masks, budgets, packing disjointness, coverage maximum, and child states. The hash binds the representation after those semantic checks.

Lemma C.1 (Soundness of a complete branch DAG). *If every reachable node satisfies its local transition rule and every leaf satisfies one of the two lower-bound rules above, then the root state has no hitting set within its budget.*

Proof. Proceed by reverse topological order. A checked leaf has no feasible hitting set by its packing or coverage inequality. At a branch node, any hitting set must select some coordinate in the named mask and would therefore induce a feasible hitting set in at least one child. If all children are infeasible, so is the parent. The conclusion reaches the root. Shared child states do not affect the induction because the graph is acyclic and every reachable node is checked once. $\square$

C.3 Complete proof census

For S_5 there are 11,893 branch nodes and 51,696 leaves. Of the leaves, 49,123 close by disjoint packing and 2573 by uniform coverage. The maximum depth is eleven. For A_5 there are 44,316 branch nodes and 169,332 leaves: 164,003 packing leaves and 5329 coverage leaves, with maximum depth fourteen. All 63,589 and 213,648 nodes are reachable from their roots.

The deterministic compressed transports and canonical uncompressed content are separately hashed:

Action	representation	SHA-256
S_5	gzip	6babe9631d4dc613...f8c60d
S_5	canonical JSON	233be7b8bb5aa9ef...200a2d
A_5	gzip	11d3efdc8b683965...d2cde
A_5	canonical JSON	866974b4163f90e3...eab82

The full hashes are recorded in the atlas manifest. The verifier checks both transport forms, but the lower-bound conclusion follows from the local rules, not from matching the displayed prefixes.

D Atlas manifest, provenance, and replay commands

This appendix is the paper-facing entry point to the certified atlas in the companion repository. The repository manifest binds every distributed source, certificate, script, and generated object by SHA-256. An event-free build attestation binds the manifest and title-named PDF without self-reference. The human-readable artifact map supplies the exact relative path for every statement in Table 7.

D.1 Principal objects

The principal five-vertex objects are the canonical four-action payload and the two deterministic-gzip lower-bound DAGs. Their complete SHA-256 digests, together with those of the four-vertex registries and certificates, appear in the repository manifest. Matching a digest establishes byte identity only; all mathematical conclusions are checked semantically.

D.2 Minimal read-only replay

From the companion-repository root, run

```
python scripts/verify.py --profile core
python -O scripts/verify.py --profile core
```

Both commands verify the manifest, rebuild the four-vertex certificate semantics, reconstruct the five-vertex coordinate data independently, and check every node of both lower-bound DAGs. They must end with

PASS_ORBIT_SUM_PACKAGE_CORE.

For a broader replay that also regenerates both four-vertex enumerations and the five-vertex payload in disposable directories, run

```
python scripts/verify.py --profile full
```

The expected endpoint is

PASS_ORBIT_SUM_PACKAGE_FULL.

No scientific output in the repository is overwritten by either profile. The mathematical scripts use only the Python standard library; PDF inspection uses the separately declared package dependency.

D.3 Table provenance

All tables are author-generated TeX constructions made only from the certified numerical summaries. They contain no third-party image asset and are cross-checked against the exact theorem summary.

Data and code availability

The proof is computer-assisted. This manuscript gives the mathematical reduction, explicit upper witnesses, and the local soundness argument for the lower-certificate checker. For the five-vertex lower bounds, the distributed pair cores and complete branch DAGs are essential finite proof objects: excluding budgets 13 and 16 requires checking all 63,589 and 213,648 reachable nodes, respectively. The designated companion repository for these materials is

<https://github.com/aconsciousfractal/FCIG-Exact-Squarefree-Orbit-Sum-Reconstruction>.

It contains the exact finite certificates and independent replay code described in Appendix D. The optimizer was used only to discover witnesses and hard pair cores; the accepted lower bounds come from the checker that reconstructs every separator mask and validates every reachable DAG transition. Third-party full texts are not redistributed.

Acknowledgment of prior frameworks

The finite-action enumeration framework is classical and is represented here by Pólya’s work. The orbit-incidence viewpoint follows the Kramer–Mesner tradition, while the language of separating invariants and reconstruction is drawn from the cited invariant-theoretic and graph-reconstruction literature. The exact four- and five-vertex numerical statements are established by the arguments and finite certificates given here; no novelty, priority, or firstness claim is made.

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